

Model Exploration and LLM Evaluation

Overview

During the development of the **Joke → Comic Generator**, multiple Large Language Models (LLMs) and image generation platforms were evaluated to determine the most suitable solution for comic-style generation. The goal was to identify a system that was:

- Cost-effective
- Capable of comic-style output
- Flexible for prompt engineering
- Deployable locally or via API
- Suitable for batch generation

Below is a summary of the tools and platforms explored.

Platforms Evaluated

1. Bing Image Creator

- Panels: 1–2
- Speed: Fast
- Cost: Free
- Limitation: Limited daily quota
- Notes: Easy to use but not suitable for programmatic batch generation or integration into a Flask-based application.

2. StableDiffusionWeb.com

- Panels: 1
- Speed: Fast
- Cost: Free
- Limitation: Browser-based only
- Notes: Useful for experimentation but lacks backend/API flexibility.

3. Comic Diffusion (Hugging Face)

- Panels: 1–Batch (GPU limited)
- Speed: Medium
- Cost: Free (HF hosting)
- Notes: Comic-style Stable Diffusion model. Can be self-hosted for unlimited generation.

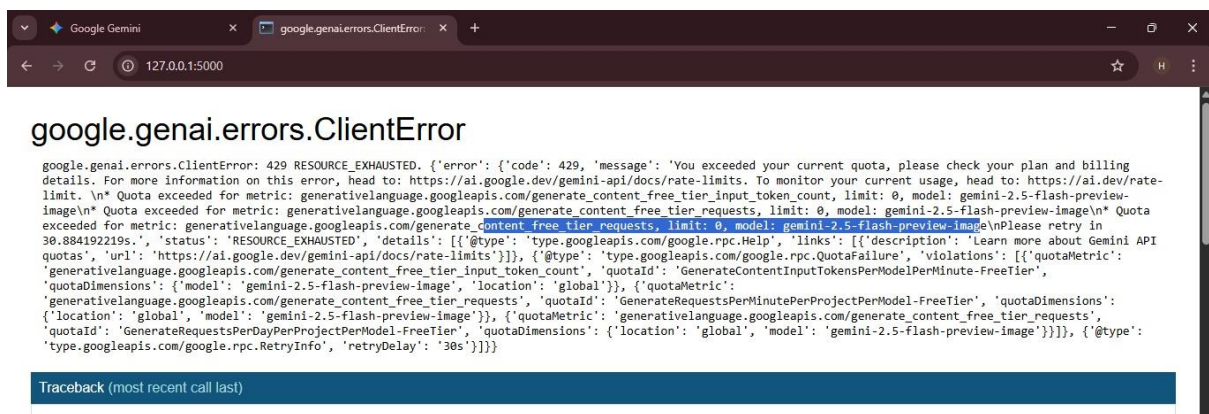
4. RetroComicFlux (Hugging Face)

- Panels: 1–Batch
- Speed: Medium
- Cost: Free
- Notes: Produces vintage comic aesthetic. Open-source but requires hosting infrastructure.

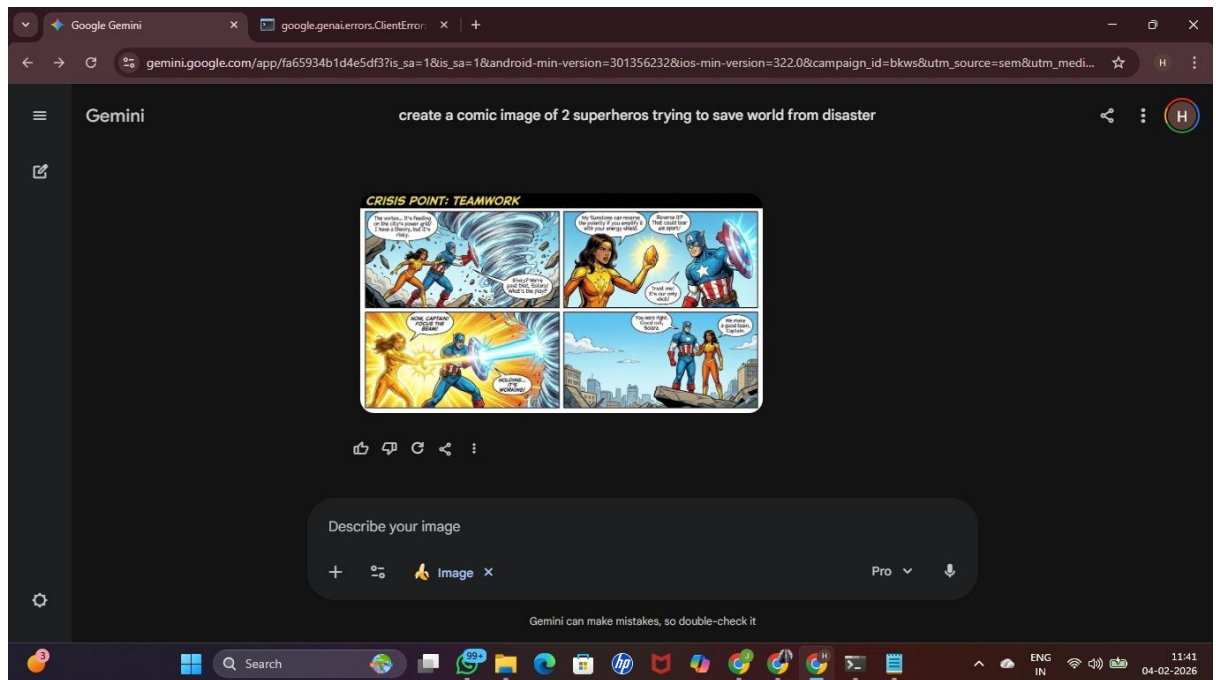
5. Craiyon (DALL·E Mini)

- Panels: 1
- Speed: Medium
- Cost: Free + Paid (\$0–\$10/month)
- Notes: Suitable for meme-style cartoons but lower image quality compared to modern diffusion models.

6. Gemini Nano Banana



- Panels: 1–3+
- Speed: Fast
- Cost: Usage-based (~\$0.039 per image)
- Notes: Google’s conversational image generation API.



- Issue: The free tier was not available for image generation API usage. Since this project required repeated testing and iteration, paid API dependency was not feasible.

Although initial implementation code was written targeting Nano Banana, lack of free-tier access prevented deployment.

7. ComicsAI / Comic-LLM

- Panels: 1–3
- Speed: Fast
- Cost: \$10–\$30/month
- Notes: Panel-based comic generator with paid tiers. Not ideal for a student project due to subscription cost.

8. AI Comic Factory (Hugging Face)

- Panels: 1

- Speed: Fast
- Cost: Free + Paid (\$0–\$35/month)
- Notes: Good comic-focused interface, but free version has strict quotas.

9. Leonardo.ai

- Panels: 1–4
- Speed: Very fast
- Cost: \$12–\$60/month
- Notes: High-quality output but token-based system limits free experimentation.

10. Flux1.ai / ComicFX / FLUX LoRA Models

- Panels: 1–Batch
- Speed: Medium
- Cost: Paid API
- Notes: High customization but not suitable without paid hosting or GPU infrastructure.

Final Model Selection

After evaluating the above options, the final implementation used:

- **GPT-2 (Hugging Face Transformers)** for comic script generation
- **Stable Diffusion v1.5 (runwayml/stable-diffusion-v1-5)** for image generation
- Hosted locally using Flask

Reason for Selection

- Fully open-source
- No API cost
- Full control over prompt engineering
- Can run locally without subscription dependency
- Scalable if deployed on GPU infrastructure

Final System Architecture

1. User inputs a joke.
2. GPT-2 generates a 3-panel comic script.
3. The generated script is passed as a prompt to Stable Diffusion.
4. Stable Diffusion generates a comic-style image.
5. Image is encoded in Base64 and displayed in the browser.

Execution Steps

Step 1: Create Virtual Environment

```
python -m venv venv
```

Activate:

```
venv\Scripts\activate
```

Step 2: Install Dependencies

```
pip install flask transformers diffusers torch accelerate safetensors
```

Step 3: Run Application

```
python app.py
```

Access via browser:

```
http://127.0.0.1:5000
```

Performance Observation

- Text generation (GPT-2): ~5–10 seconds
- Image generation (Stable Diffusion on CPU): ~4–5 minutes per comic

Total Time Per Comic:

Approximately **5 minutes** on CPU-based system without GPU acceleration.

If run on CUDA-enabled GPU:

- Generation time reduces to ~20–40 seconds per comic.

Limitations

- High RAM usage (~8GB+ recommended)
- Large model download (~4GB for Stable Diffusion v1.5)
- Slow generation on CPU
- Single-image output (not separate panel rendering)

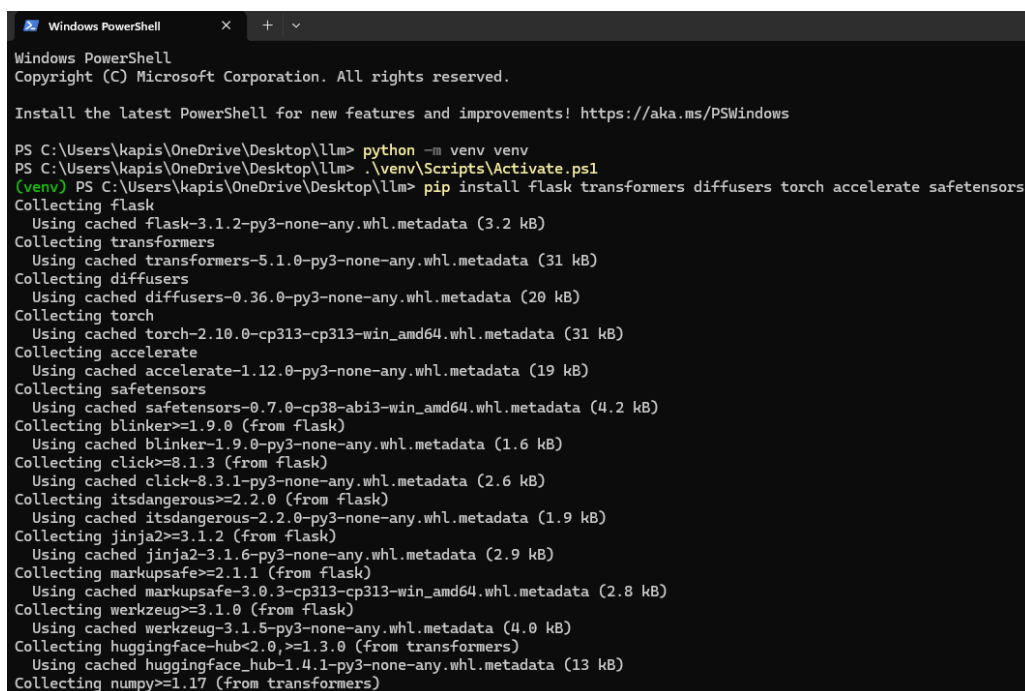
Conclusion

Although several commercial and hosted comic-generation platforms were explored, the final implementation prioritizes:

- Cost-free deployment
- Open-source architecture
- Full backend control
- Educational value in understanding LLM + Diffusion pipelines

The current system successfully converts user jokes into AI-generated comic visuals, with an average generation time of approximately five minutes per comic on CPU.

Steps In windows PowerShell:



```
Windows PowerShell
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Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\kapis\OneDrive\Desktop\llm> python -m venv venv
PS C:\Users\kapis\OneDrive\Desktop\llm> .\venv\Scripts\Activate.ps1
(venv) PS C:\Users\kapis\OneDrive\Desktop\llm> pip install flask transformers diffusers torch accelerate safetensors
Collecting flask
  Using cached flask-3.1.2-py3-none-any.whl.metadata (3.2 kB)
Collecting transformers
  Using cached transformers-5.1.0-py3-none-any.whl.metadata (31 kB)
Collecting diffusers
  Using cached diffusers-0.36.0-py3-none-any.whl.metadata (20 kB)
Collecting torch
  Using cached torch-2.10.0-cp313-cp313-win_amd64.whl.metadata (31 kB)
Collecting accelerate
  Using cached accelerate-1.12.0-py3-none-any.whl.metadata (19 kB)
Collecting safetensors
  Using cached safetensors-0.7.0-cp38-abi3-win_amd64.whl.metadata (4.2 kB)
Collecting blinker>=1.9.0 (from flask)
  Using cached blinker-1.9.0-py3-none-any.whl.metadata (1.6 kB)
Collecting click>=8.1.3 (from flask)
  Using cached click-8.3.1-py3-none-any.whl.metadata (2.6 kB)
Collecting itsdangerous>=2.2.0 (from flask)
  Using cached itsdangerous-2.2.0-py3-none-any.whl.metadata (1.9 kB)
Collecting jinja2>=3.1.2 (from flask)
  Using cached jinja2-3.1.6-py3-none-any.whl.metadata (2.9 kB)
Collecting markupsafe>=2.1.1 (from flask)
  Using cached markupsafe-3.0.3-cp313-cp313-win_amd64.whl.metadata (2.8 kB)
Collecting werkzeug>=3.1.0 (from flask)
  Using cached werkzeug-3.1.5-py3-none-any.whl.metadata (4.0 kB)
Collecting huggingface-hub<2.0,>=1.3.0 (from transformers)
  Using cached huggingface_hub-1.4.1-py3-none-any.whl.metadata (13 kB)
Collecting numpy>=1.17 (from transformers)
```

```
Using cached annotated_doc-0.0.4-py3-none-any.whl (5.3 kB)
Installing collected packages: mpmath, zipp, urllib3, typing-extensions, sympy, shellingham, setuptools, safetensors, regex, pyyaml, psutil, Pillow, packaging, numpy, networkx, markupsafe, itsdangerous, idna, hf-xet, h11, fsspec, filelock, colorama, charset-normalizer, certifi, blinker, annotate
d-doc, werkzeug, tqdm, requests, jinja2, importlib_metadata, httpcore, click, anyio, typer-slim, torch, httpx, flask, huggingface-hub, tokenizers, d
iffusers, accelerate, transformers
Successfully installed Pillow-12.1.1 accelerate-1.12.0 annotated-doc-0.0.4 anyio-4.12.1 blinker-1.9.0 certifi-2026.1.4 charset-normalizer-3.4.4 clic
k-8.3.1 colorama-0.4.6 diffusers-0.36.0 filelock-3.20.3 flask-3.1.2 fsspec-2026.2.0 h11-0.16.0 hf-xet-1.2.0 httpcore-1.0.9 httpx-0.28.1 huggingface-
hub-1.4.1 idna-3.11 importlib_metadata-8.7.1 itsdangerous-2.2.0 jinja2-3.1.6 markupsafe-3.0.3 mpmath-1.3.0 networkx-3.6.1 numpy-2.4.2 packaging-26.0
psutil-7.2.2 pyyaml-6.0.3 regex-2026.1.15 requests-2.32.5 safetensors-0.7.0 setuptools-82.0.0 shellingham-1.5.4 sympy-1.14.0 tokenizers-0.22.2 torc
h-2.10.0 tqdm-4.67.3 transformers-5.1.0 typer-slim-0.21.2 typing-extensions-4.15.0 urllib3-2.6.3 werkzeug-3.1.5 zipp-3.23.0

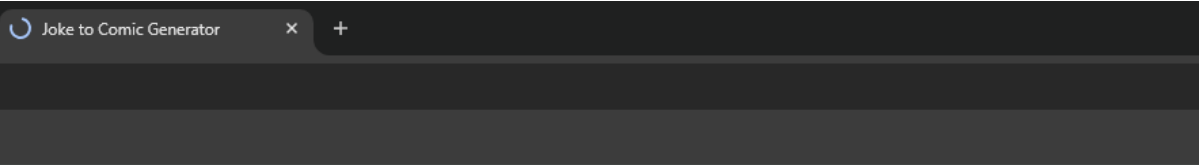
[notice] A new release of pip is available: 25.1.1 -> 26.0.1
[notice] To update, run: python.exe -m pip install --upgrade pip
(venv) PS C:\Users\kapis\OneDrive\Desktop\llm> python app.py
C:\Users\kapis\OneDrive\Desktop\llm\venv\Lib\site-packages\diffusers\models\transformers\transformer_kandinsky.py:168: UserWarning: CUDA is not avai
lable or torch_xla is imported. Disabling autocast.
  @torch.autocast(device_type="cuda", dtype=torch.float32)
C:\Users\kapis\OneDrive\Desktop\llm\venv\Lib\site-packages\diffusers\models\transformers\transformer_kandinsky.py:272: UserWarning: CUDA is not avai
lable or torch_xla is imported. Disabling autocast.
  @torch.autocast(device_type="cuda", dtype=torch.float32)
Loading weights: 100%|██████████| 148/148 [00:00<00:00, 420.93it/s, Materializing param=transformer.wte.weight]
GPT2LMHeadModel LOAD REPORT from: gpt2
Key | Status | |
-----|-----|
h.{0...11}.attn.bias | UNEXPECTED | |

Notes:
- UNEXPECTED :can be ignored when loading from different task/architecture; not ok if you expect identical arch.
Loading weights: 100%|██████████| 396/396 [00:00<00:00, 1100.41it/s, Materializing param=visual_projection.weight]
StableDiffusionSafetyChecker LOAD REPORT from: C:\Users\kapis\.cache\huggingface\hub\models--runwayml--stable-diffusion-v1-5\snapshots\451f4fe16113b
ff5a5d2269ed5ad43b0592e9a14\safety_checker
Key | Status | |
-----|-----|
vision_model.vision_model.embeddings.position_ids | UNEXPECTED | |
```

```
Notes:
- UNEXPECTED :can be ignored when loading from different task/architecture; not ok if you expect identical arch.
Loading weights: 100%|██████████| 196/196 [00:00<00:00, 1058.37it/s, Materializing param=text_model.final_layer_norm.weight]
CLIPTextModel LOAD REPORT from: C:\Users\kapis\.cache\huggingface\hub\models--runwayml--stable-diffusion-v1-5\snapshots\451f4fe16113bff5a5d2269ed5ad
43b0592e9a14\text_encoder
Key | Status | |
-----|-----|
text_model.embeddings.position_ids | UNEXPECTED | |

Notes:
- UNEXPECTED :can be ignored when loading from different task/architecture; not ok if you expect identical arch.
Loading pipeline components...: 100%|██████████| 7/7 [00:02<00:00, 3.28it/s]
* Serving Flask app 'app'
* Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on http://127.0.0.1:5000
Press CTRL+C to quit
* Restarting with stat http://127.0.0.1:5000
C:\Users\kapis\OneDrive\Desktop\llm\venv\Lib\site-packages\diffusers\models\transformers\transformer_kandinsky.py:168: UserWarning: CUDA is not avai
lable or torch_xla is imported. Disabling autocast.
  @torch.autocast(device_type="cuda", dtype=torch.float32)
C:\Users\kapis\OneDrive\Desktop\llm\venv\Lib\site-packages\diffusers\models\transformers\transformer_kandinsky.py:272: UserWarning: CUDA is not avai
lable or torch_xla is imported. Disabling autocast.
  @torch.autocast(device_type="cuda", dtype=torch.float32)
Loading weights: 100%|██████████| 148/148 [00:00<00:00, 245.00it/s, Materializing param=transformer.wte.weight]
GPT2LMHeadModel LOAD REPORT from: gpt2
Key | Status | |
-----|-----|
h.{0...11}.attn.bias | UNEXPECTED | |

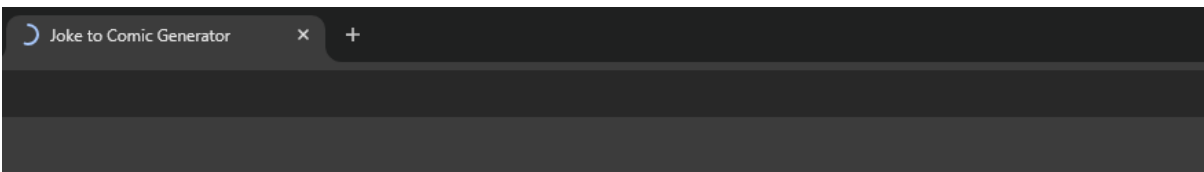
Notes:
- UNEXPECTED :can be ignored when loading from different task/architecture; not ok if you expect identical arch.
Loading weights: 100%|██████████| 396/396 [00:01<00:00, 253.49it/s, Materializing param=visual_projection.weight]
StableDiffusionSafetyChecker LOAD REPORT from: C:\Users\kapis\.cache\huggingface\hub\models--runwayml--stable-diffusion-v1-5\snapshots\451f4fe16113b
```



Joke → Comic Generator

Enter your joke here...

Generate Comic



Joke → Comic Generator

a cat flying over the city

Generate Comic

```
Notes:
- UNEXPECTED :can be ignored when loading from different task/architecture; not ok if you expect identical arch.
Loading pipeline components...: 100%| 7/7 [00:05<00:00, 1.38it/s]
* Debugger is active!
* Debugger PIN: 230-978-651
127.0.0.1 -- [11/Feb/2026 13:34:39] "GET / HTTP/1.1" 200 -
127.0.0.1 -- [11/Feb/2026 13:34:39] "GET /favicon.ico HTTP/1.1" 404 -
Passing 'generation_config' together with generation-related arguments({'max_length', 'do_sample'}) is deprecated and will be removed in future versions. Please pass either a 'generation_config' object OR all generation parameters explicitly, but not both.
Setting 'pad_token_id' to 'eos_token_id':50256 for open-end generation.
Both 'max_new_tokens' (=256) and 'max_length' (=200) seem to have been set. 'max_new_tokens' will take precedence. Please refer to the documentation for more information. (https://huggingface.co/docs/transformers/main/en/main_classes/text_generation)
Token indices sequence length is longer than the specified maximum sequence length for this model (245 > 77). Running this sequence through the model will result in indexing errors
The following part of your input was truncated because CLIP can only handle sequences up to 77 tokens: ["was a joke comic : a cat flying over the city joke : well , this is not going to work comic : a cat flying over the city comic : a cat flying over the city joke : yeah , well , i 'm pretty sure that was a joke comic : a cat flying over the city comic : a cat flying over the city joke : no , it 's just a joke joke : okay , so when the cat dies , you 're going to die comic : a cat flying over the city comic : a cat flying over the city comic : a cat flying over the city comic : a cat flying over the city joke : oh , i 'm not going to die comic : a cat flying over the city comic : a cat flying over the city comic : a cat flying over the city comic : a cat flying over the city joke : ok , so"]
82%| 41/50 [24:38<03:46, 25.15s/it]
```

Generated Comic Image:

